

Portfolio Case Study — WFP Kano Food Security Dashboard

Project Overview

I developed a humanitarian analytics dashboard inspired by the World Food Programme (WFP) to monitor food security conditions across selected Local Government Areas (LGAs) in Kano State, Nigeria.

This project was designed to simulate a real-world decision-support system used by humanitarian organizations to identify hunger hotspots, assess household vulnerability, and track market-driven food insecurity.

The solution transforms raw household survey data into actionable insights for operational planning and intervention prioritization.

Business Problem

Food insecurity remains a critical challenge in many communities due to:

- * Rising food inflation
- * Low household income
- * Large family sizes
- * Economic vulnerability
- * Displacement and instability

Humanitarian agencies require timely, reliable, and visual reporting tools to answer key operational questions:

- * Which areas are most vulnerable?
- * Which households face severe hunger?

- * How are market prices affecting purchasing power?
- * Where should aid be prioritized?

Without centralized analytics, reporting becomes fragmented and slow.

Project Objectives

The primary objectives of this project were to:

- * Monitor household food security indicators
- * Identify high-risk hunger zones
- * Analyze food market inflation trends
- * Measure vulnerability across LGAs
- * Support evidence-based intervention planning

Dataset Development

Since no public Kano-specific WFP dataset was available, I generated a realistic synthetic dataset using Python.

The dataset includes over **3,000 household records** with:

- * Household demographics
- * Income sources and monthly income
- * Household size
- * Gender of household head

- * Food Consumption Score (FCS)
- * Reduced Coping Strategy Index (rCSI)
- * Hunger classification
- * LGA segmentation
- * Market price trends for staple foods

Data was cleaned and validated in Microsoft Excel before being modeled in Power BI.

Tools & Technologies

- * Python (Synthetic Data Generation)
- * Microsoft Excel (Data Cleaning & Validation)
- * Power BI (Dashboard Development)
- * DAX (Measures & KPIs)

Dashboard Features

The dashboard consists of four interactive pages:

1. Executive Summary

Provides high-level KPIs:

- * Total Households Surveyed
- * Average Household Income
- * Severe Hunger %

- * Critical Hunger %

- * Average FCS

- * Average rCSI

2. Food Risk Analysis

Analyzes:

- * Household size vs hunger severity

- * Income vs vulnerability

- * Gender-based risk patterns

- * IDP impact

3. Market Monitoring

Tracks:

- * Rice price trends

- * Millet price trends

- * Food inflation by LGA

- * Purchasing pressure

4. Insights & Recommendations

Summarizes major findings and operational recommendations.

Key Insights

Analysis revealed several important findings:

- * Rural LGAs showed higher food insecurity than urban LGAs
- * Low-income households faced significantly higher hunger risk
- * Larger households (8+ members) were more vulnerable
- * Food inflation reduced household purchasing power
- * Households with low FCS had higher coping stress

Business Impact

This dashboard enables humanitarian stakeholders to:

- * Detect hunger hotspots faster
- * Prioritize intervention areas
- * Improve aid allocation efficiency
- * Support donor reporting
- * Enhance evidence-based planning

The project demonstrates how analytics can improve humanitarian decision-making by converting complex survey data into clear operational intelligence.

Skills Demonstrated

This project showcases my ability to:

- * Design end-to-end analytics solutions

- * Generate realistic datasets using Python
- * Clean and validate data
- * Build professional Power BI dashboards
- * Apply KPI-driven storytelling
- * Communicate business insights effectively

Final Outcome

The final deliverables include:

- * Python data generation script
- * Excel dataset
- * Power BI dashboard
- * Executive PDF report
- * Presentation deck
- * Portfolio-ready documentation

This project reflects the type of analytical work required in organizations such as WFP, UNICEF, UNDP, and other humanitarian agencies.